

Md. Parvej Hoque Palash

✉ mdparvejhoquepalash.official@gmail.com | 📞 +88 01841-763871
🏠 [Portfolio](#) | 🌐 [Parvej-Hoque-Palash](#) | 🌐 [parvej-hoque-palash](#)

About Me

Software Engineer with experience in data-driven application development, database design, reporting systems, and business analytics solutions. Skilled in translating business requirements into scalable data models, reports, dashboards, and decision-support tools. Proficient in Python, SQL, PostgreSQL, SQLAlchemy, Power BI, and data processing techniques, with hands-on experience in query optimization, reporting automation, and business intelligence development. Strong ability to collaborate with stakeholders to deliver actionable insights, improve operational efficiency, and support data-driven decision-making.

Experience

Divine IT Limited | Software Engineer

April 2025 - Present

- Developed and maintained enterprise applications, reporting systems, and data-driven solutions using Python, SQLAlchemy, PostgreSQL, and JavaScript.
- Collaborated with business stakeholders, analysts, QA teams, and clients to gather requirements and translate them into scalable data models, reports, dashboards, and software solutions.
- Designed and generated business reports, regulatory reports, KPI dashboards, and visualization components to support operational monitoring and strategic decision-making.
- Built and optimized complex SQL queries and database structures to improve reporting performance, data accuracy, and system efficiency.
- Developed VAT calculation modules, Mushak reports, and analytics dashboards within PrismVAT, enabling automated compliance reporting and business performance tracking.
- Designed and implemented an Auction (Bidding) Module in PrismERP, providing real-time monitoring, workflow automation, sales tracking, and operational reporting capabilities.
- Enhanced third-party integrations and notification services while improving application scalability and data flow reliability.
- Delivered new functionalities and reporting features aligned with evolving business requirements while maintaining system stability and performance.

Divine IT Limited | Trainee Software Engineer

July 2024 - March 2025

- Developed a SQL Query Builder Editor using React and Redux, enabling users to generate dynamic queries and perform data analysis efficiently.
- Designed, optimized, and refactored over 1,000 SQLAlchemy queries during migration from MySQL to PostgreSQL, improving query performance, maintainability, and reporting efficiency.
- Implemented dashboard management and data access features, including configurable reporting components, session-based search management, and role-based access control.
- Performed query tuning, testing, debugging, and performance optimization to ensure reliable reporting and accurate business data delivery.
- Worked closely with cross-functional teams to resolve data-related issues and improve overall application and reporting performance.

Education

Jahangirnagar University (JU)

Graduation date: May, 2024

Completed **B.Sc in Computer Science and Engineering** | [CSE-JU](#)

CGPA: 3.57/4.00

Relevant Courses: Data Structure, Algorithms I, Algorithms II, OOP (C++), OS, SWE, DTC, SQA, Web, ML, AI etc.

Jahangirnagar University (JU)

Graduation date: December, 2025

Completed **M.Sc in Computer Science and Engineering** | [CSE-JU](#)

CGPA: 3.53/4.00

Projects

Albab Academy

React, Nextjs, PostgreSQL, Tailwind, Github

- Developed a comprehensive school management platform featuring online admissions, payment automation, student information management, and report generation.
- Designed database structures and administrative dashboards to support operational monitoring and data-driven decision-making. [🔗Live](#)

Library Management

React, Nodejs, MongoDB, Tailwind, Github

- Developed an administrative dashboard for monitoring library operations and visualizing book statistics across multiple categories.
- Implemented reporting and analytics features to provide insights into inventory utilization and resource distribution. [🔗Github](#)

Employee Management

React, Nodejs, MongoDB, Tailwind, Github

- Built a centralized employee management platform with role-based access control and CRUD operations for workforce data management.
- Designed database workflows and user management features to improve data accessibility and administrative efficiency. [🔗Github](#) [👁️Live](#)

Genomedics

React, Nextjs, PostgreSQL, Tailwind, Github

- Developed and maintained a responsive web platform for a pharmaceutical supplier, ensuring efficient content management and data accessibility.
- Implemented database-driven functionalities and optimized data retrieval processes to enhance application performance. [👁️Live](#)

Research Works

DDoS Attack Detection Using Machine Learning

Google Colab, Overleaf, Github

- It is a machine learning-based project for detecting DDoS attacks using machine learning algorithms. I used **SDN dataset** in this project. I have used five machine learning models and find out the best-performing model among them. [🔗Github](#) [📄Overleaf](#)

A Deep Learning Approach for Parkinson's Disease Progression Forecasting

Kaggle, Google Colab, Overleaf

- It is a deep learning-based project for forecasting Parkinson's Disease using LSTM and Transformer models. I used **NTUA Parkinson** time-series dataset in this project. [📄Overleaf](#)

Programming Competitions & Achievements

- Participated in the **ICPC Preliminary Contest** twice.
- Participated in **100+** online contests in different judges.
- Solved **1000+** programming problems in different online judges.
- Ranked **29th out of 511 teams** in **CODE SAMURAI 2022 Hackathon**.
- Ranked **19th out of 400 teams** in **CODE SAMURAI 2024 Hackathon**.
- Got scholarship **3 times** out of 4 years for good results while studying B.Sc.

Online Judge Handles: **Leetcode:** [php_980](#) **Codeforces:** [php_980](#) **Vjudge:** [palash_28_378](#)

Courses & Certificates

- **Bootcamp: MERN & PERN Full Stack** from [BongoDev](#) *Dec. 2022 - July. 2023*
- **Masterclass: Business Analytics with PowerBI** from [Unbolt Academy](#) *April 2026*

Technical Skills

Languages: Python, SQL, JavaScript, C, C++, HTML, CSS

Data & Analytics: PostgreSQL, MySQL, Redshift, SQLAlchemy, Pandas, Power BI, Data Modeling, Query Optimization, Reporting & Dashboard Development, Business Intelligence & KPI Reporting, Data Analysis & Visualization

Frameworks & Databases: Django, React, Node.js, Express.js, Next.js, PostgreSQL, MySQL, MongoDB

Technologies & Tools: REST API, Eclipse (BIRT Reporting), GitHub, Postman, Linux Environment, Figma, Cloudflare, Vercel, Netlify